Vidit Agarwal

San Jose, CA | (408) 332-4884 | vidit3035@gmail.com | linkedin.com/in/viditagarwal- | github.com/v1dit | v1dit.github.io

Education

University of California, San Diego | Computer Science & Mathematics — Expected June 2028

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Machine Learning, Cloud Computing.

Experience

PhrasIQ | *Software Engineering Intern, Enterprise finance data & AI* — Jul 2026 – Present

- Built **production Python/FastAPI backend services and REST APIs** for CRM-style financial data management and analytics workflows with founding engineers.
- Integrated those APIs into **customer-facing finance dashboards** that surface customer records and financial metrics for review in one place.

Circuit Launch | *Full-Stack Engineering Intern, Physical intelligence & robotics startup hub* — Jun 2026 – Sep 2026

- Built backend workflows for **Solo Tech's physical AI data-collection hub** to capture and organize new robotic-arm movements across different arm types.

Unicage | *Blockchain Developer, Blockchain transaction infrastructure* — Jun 2025 – May 2026

- Built backend APIs and automation for UK client transaction platforms, supporting **5,000+ decentralized transactions**.
- Integrated payment and logging services with real-time data pipelines for transaction tracking across SaaS products.

Santa Clara University | *Lead Lab Researcher, Open-space modeling & geospatial ML* — Jan 2025 – Jun 2026

- Built a **statewide open-space modeling pipeline** with Google Earth Engine and LightGBM to analyze California land-cover change from 1984 to 2024.
- Combined Landsat, MODIS, and AlphaEarth embeddings across 7 land-cover classes; coauthored a **GISTAM 2026 paper** reporting **13.2% statewide loss** and $R^2 = 0.954$ on **14,309 samples**.
- Improved multimodal model performance by **15%** through evaluation and tuning of visual and text-based data pipelines.

Berkeley Lab | *Data Analyst Intern, Rainforest & watershed remote sensing* — Jun 2022 – Aug 2023

- Built automated pipelines to collect satellite imagery and calculate NDVI features from rainforest and watershed study areas, producing model-ready geospatial datasets.
- Trained geospatial models on those datasets to **predict vegetation health and environmental change** across watersheds.

Technical Skills

Languages: Java, Python, JavaScript/TypeScript, Go, C++, SQL, HTML/CSS
Frameworks: React, Next.js, FastAPI, Node.js, Fastify, Pydantic, scikit-learn
Data & AI: pandas, NumPy, SQLite, LightGBM, Google Earth Engine, vector search, RAG, MCP
Systems & Tools: Git/GitHub, GitHub Actions, Docker, Kubernetes, Linux, AWS, Google Cloud, Vitest

Projects

Ido | *Python, FastAPI, Pydantic, Blender, OpenSCAD*

- Built a **universal 3D/CAD agent harness** that turns prompts into validated engineering IR and editable models through an engine-agnostic adapter architecture, with Blender and OpenSCAD integrations.

Kiro | *Next.js, Fastify, MCP, Hyperspell, Nia*

- Built a **company-memory system for coding agents** that cross-references internal decisions with code, issues, and PRs; MCP retrieves task-relevant context and flags conflicts. Demo guided a new hire's **first PR in under 20 minutes**.

Wooly | *Python, speech-to-text, LLM function calling, mobile automation*

- Built **computer use for phones**: converted voice requests into validated, executable actions for messaging, calls, navigation, and multi-step mobile workflows.

ML Labs | *Next.js, Node.js, Python, pandas, scikit-learn*

- Built an **end-to-end autonomous ML lab** with **10 specialized agents** that take a dataset through feature engineering, model selection, training, evaluation, paper drafting, and a prediction API.

Consulting

DS3 Consulting | **Current Client: JPMorgan** | *Software Engineering Consultant* — Fall 2026 – Present

Awards & Involvement

Hackathons | **$8K+ earned across 6 events: 1st Place** – Anthropic x AWS (Ido), ML Empowerment Foundation Build Challenge (ML Labs), YC x Nozomio Convex Track (Kiro), OceanHacks (BlueDocs); **2nd Place** – AWS Building the Next YC Unicorn; **3rd Place** – SCU ACM Hackathon.
Open Source: Shipped backend microservices for the Linux Foundation's mobile-core project and landed merged PRs across other organizations' open-source codebases.
Clubs: Sangam, Alpha Phi Omega (APO), Association for Computing Machinery (ACM).